

- **SYSTEM** - QUANTITY OF MATTER
- **CLOSED SYSTEM** - FIXED QUANTITY OF MASS
- **OPEN SYSTEM** (CONTROL VOL) - FIXED VOLUME TO STUDY
- ALL SYSTEMS HAVE A BOUNDARY - SURROUNDING ENVIRONMENT



HEAT AND

WORK ARE ENERGY TRANSFERS

- **SIMPLE COMPRESSIBLE SYSTEMS** - IGNORE MANY OTHER MECHANICAL PROPERTIES; ELECTRICAL, MAGNETIC, GRAVITATIONAL
- **ENERGY** - ABILITY TO PERFORM WORK OR EXCHANGE HEAT ENERGY CAN BE STORED TO OUR SYSTEM, OR IT CAN BE TRANSFERRED (WORK)

- ANY SUBSTANCE WILL HAVE A COLLECTION OF PROPERTIES (IDENTIFIED, MEASURED OR CALCULATED)

- A SUBSTANCE SET OF PROPERTIES IS CALLED ITS STATE THE STATE OF A SIMPLE COMPRESSIBLE SYSTEM CAN BE DETERMINED BY KNOWING ANY 2 INDEPENDENT INTENSIVE PROPERTIES

- **INTENSIVE PROPERTY** - DOES NOT DEPEND ON ITS SIZE - TEMP, PRESSURE, DENSITY

- **EXTENSIVE PROPERTY** - DEPENDENT ON SIZE

- **VOLUME, INTERNAL ENERGY (U)**

SPECIFIC PROPERTY - EXTENSIVE PROPERTY, PER UNIT MASS

($\frac{U}{m}$ - SPECIFIC ENERGY) / ($\frac{V}{m}$ - SPECIFIC VOLUME)

EXTENSIVE PROPERTIES MORE EXTENSIVE ON MORE MEASURABLE PROPERTIES MEASUREMENT PROPERTIES

- FINDING 2 INDEPENDENT PROPERTIES ALLOW US TO USE TABLES TO FIND OTHER PROPERTIES OF THE SUBSTANCE IN THAT STATE

- **EQUILIBRIUM** - IF INTENSIVE PROPERTIES DO NOT CHANGE OVER TIME, 1) MECHANICAL EQUILIBRIUM - NO UNBALANCED FORCES

- 2) THERMAL EQUILIBRIUM - NO TEMPERATURE DIFFERENCES

- 3) THERMODYNAMIC EQUILIBRIUM - BOTH MECHANICAL AND THERMAL EQUILIBRIUM

PROCESS - IF A PROPERTY CHANGES WHEN ITS STATE CHANGES, AND A PROCESS HAS OCCURRED, WE TALK ABOUT CHANGE PROCESSES OVER VERY SHORT, SO WE ARE ALWAYS WORKING AT EQUILIBRIUM (QUASI-EQUILIBRIUM PROCESSES), IF A PROPERTY REMAINS CONSTANT DURING A PROCESS WE USE "IS" - ISOTHERMAL (T), ISOBARIC (P), ISOCAL (Q) - PROCESS THAT OCCURS AND STAYS AT SAME STATE

STEADY-STATE PROCESSES - DOES NOT CHANGE OVER TIME (OPEN SYSTEM)

UNITS: MASS - kg, LENGTH - m, TIME - s, PRESS - Pa (N/m²)

HEAT (Q) - J, W - J/s

WALL - Pa, LENGTH - m, TIME - s, PRESS - Pa (N/m²)

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- **ENERGY TRANSFERS** - HEAT OR WORK

- **HEAT** - TRANSFERRED AT THE BOUNDARY, IT IS NOT A PROPERTY

HEAT IS POSITIVE IF ADDED TO THE SYSTEM. $Q = \text{HEAT [kJ]}$

$\frac{Q}{m} = q = \text{HEAT PER UNIT MASS [kJ/kg]}$

$\frac{Q}{\dot{m}} = \dot{q} = \text{HEAT TRANSFER RATE [kW]} \quad \left[\frac{\text{kJ}}{\text{s}} \right]$

PROCESS FROM 1 TO 2: Q_{12} (USE SUBSCRIPTS)

- **WORK** - ENERGY TRANSFER AT THE BOUNDARY, NOT A PROPERTY.

WORK IS POSITIVE IF IT IS DONE BY SYSTEM ON THE SURROUNDINGS.

WORK (W) = $\int \mathbf{F} \cdot d\mathbf{s}$ (MECHANICS)

$\frac{W}{m} = w = \text{WORK PER UNIT MASS} \quad \frac{W}{\dot{m}} = \dot{w} = \text{POWER [kW]} \quad \left[\frac{\text{kJ}}{\text{s}} = \text{kW} \right]$

HEAT TRANSFER AND WORK ARE CONSIDERED PATH FUNCTIONS

POINT FUNCTIONS ARE DIFFERENTIABLE (U), PATH FUNCTIONS ARE DEPENDANT ON THE PATH.

- **1ST LAW** - CONSERVATION OF ENERGY

$E_{in} - E_{out} + E_{system} = \Delta E = E_2 - E_1$

$E = U + KE + PE \quad \therefore \Delta E = \Delta U + \Delta KE + \Delta PE$ THESE 2 EQUATIONS ARE EQUAL.

BUT $E_{in} - E_{out} = (U_{in} - U_{out}) + (KE_{in} - KE_{out}) + (PE_{in} - PE_{out})$

BETTER WRITTEN: $-W + Q + \Delta E_{flow}$

Q) $E_{in} = \dot{m} \cdot h_{in}$

$Q_{in} = 30 \text{ kJ/s}$ $Q_{out} = 5 \text{ kJ/s}$

$W_{out} = 500 \text{ W}$

$E_2 - E_1 = Q - W + \Delta E_{flow}$

$E_2 - 10 \text{ kJ/s} = (30 \text{ kJ/s} - 5 \text{ kJ/s}) - (-0.05 \text{ kJ/s}) + 0$

$E_2 = 35.5 \text{ kJ/s}$

Q) $Q_{in} = 20 \text{ kJ/s}$

$Q_{out} = 10 \text{ kJ/s}$ $W_{out} = 10 \text{ kJ/s}$

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Physical Quantities					
	Quantity	Definition	Formula	Units	Dimensions
Basic Mechanical	Length or Distance	fundamental	l	m (meter)	L (length)
	Time	fundamental	t	s (second)	T (time)
	Mass	fundamental	m	kg (kilogram)	M (mass)
	Area	distance ²	$A = l^2$	m ²	L ²
	Volume	distance ³	$V = l^3$	m ³	L ³
	Density	mass / volume	$\rho = m/V$	kg/m ³	M/L ³
	Velocity	distance / time	$v = dl/dt$	m/s	L/T
	Acceleration	velocity / time	$a = dv/dt$	m/s ²	L/T ²
	Momentum	mass × velocity	$p = mv$	kg m/s	ML/T
	Force	mass × acceleration	$F = ma$	N (newton) = kg m/s ²	ML/T ²
Mechanical	Weight	mass × acceleration of gravity	$W = mg$		
	Pressure or Stress	force / area	$P = F/A$	Pa (pascal) = N/m ² = kg/m ² /s ²	M/L ² /T ²
	Work	force × distance	$W = Fd$	J (joule) = N m = kg m ² /s ²	M L ² /T ²
	Kinetic Energy	mass × velocity ² / 2	$KE = mv^2/2$		
	Potential Energy	mass × acceleration of gravity × height			
	Power	energy / time	$P = E/t$	W (watt) = J/s = kg m ² /s ³	M L ² /T ³
	Impulse	force × time	$S = Ft$	J s = kg m ² /s ²	M L ² /T ²
	Momentum	momentum × distance	$S = p d$	(h) (quanta of action)	
	Angle	fundamental	θ	dimensionless; rad (radian); rev (revolution)	
	Rotational Mechanical	Cycles	fundamental	n	cycle (cycles)
Angular Velocity		angle / time	$\omega = d\theta/dt$	Hz (hertz) = cycles / s = 1/s	1/T
Angular Acceleration		angular velocity / time	$\alpha = d\omega/dt$	rad/s ² = 1/s ²	1/T ²
Moment of Inertia		mass × radius ²	$I = mr^2$	kg m ²	M L ²
Angular Momentum		radius × momentum	$L = rp$	J s = kg m ² /s	M L ² /T
Torque or Moment		radius × force	$\tau = rf$	(h) (quanta of angular momentum)	
Angular Velocity		moment of inertia	$\omega = L/I$		
Angular Acceleration		angular acceleration	$\alpha = d\omega/dt$		
Temperature		fundamental	T	°C (Celsius), K (Kelvin)	K (temp.)
Thermal		Heat	heat energy	Q	J (joule) = kg m ² /s ²
	Entropy	heat / temperature	$S = Q/T$	J/K	M L ² /T ² K
	Electric Charge +/-	fundamental	q	C (coulomb)	Q (charge)
	Current	charge / time	$i = q/t$	A (ampere) = C/s	Q/T
	Voltage or Potential	energy / charge	$V = Eq$	V (volt) = J/C	M L ² /Q T ²
	Resistance	energy / current	$R = V/i$	Ω (ohm) = V/A	M L ² /Q T ²
	Capacitance	charge / voltage	$C = q/V$	F (farad) = C/V	Q ² /M L ²
	Inductance	voltage / (current / time)	$L = V/i dt$	H (henry) = V s/A	M L ² /Q T ²
	Electric Field	voltage / distance	$E = V/d$	V/m = N/C	M L ² /Q T ²
	Electromagnetic	Electric Charge	electric field × area	$Q = EA$	V m = N m ² /C
Magnetic Flux		force / (charge × velocity)	$\Phi_B = F/qv$	(T) (tesla) = Wb/m ² = N s/(C m)	
Induced Voltage		change in magnetic flux / time	$\mathcal{E} = d\Phi_B/dt$	V (volt)	
Induced Current		induced voltage / resistance	$i = \mathcal{E}/R$	A (ampere)	

T-V, P-V Diagram, and properties example video